

From Pixels to Plans: Training VLA Agents via Curriculum Reinforcement Learning

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Abstract—Recent advances in Vision-Language Models (VLMs) have demonstrated impressive capabilities in semantic understanding, but often lack the ability to execute complex, multi-stage physical tasks. This work presents a Vision-Language-Action (VLA) agent capable of solving partially observable, sparse-reward navigation puzzles from raw pixel inputs. We propose a specialized architecture that fuses a frozen CLIP text encoder with a custom visual encoder utilizing Spatial Softmax to extract precise coordinate features. To address the challenge of learning long-horizon dependencies in environments with sparse rewards, we introduce a curriculum learning framework augmented by dynamic reward shaping. Our approach demonstrates successful transfer learning, where an agent pre-trained on primitive physics tasks adapts to a complex 2D obstructed maze, achieving a rollout success rate of approximately 60%. These results highlight the effectiveness of combining structured curricula with neuro-symbolic visual abstractions for training robust robotic agents.

Index Terms—vision-language-action models, curriculum reinforcement learning, sparse rewards, spatial softmax, object permanence

I. INTRODUCTION

Embodied agents must perceive their environment, interpret task objectives, and execute long-horizon action sequences under partial observability. Reinforcement learning (RL) has demonstrated success in learning control policies directly from visual observations, enabling agents to operate from raw pixel inputs without hand-crafted state representations [1]. However, pixel-based RL remains challenging in environments with sparse rewards, delayed credit assignment, and complex dependency chains, where unguided exploration often fails to discover successful behaviors.

Recent advances in Vision-Language Models (VLMs) have shown that large-scale language supervision can provide powerful semantic grounding for visual representations. Models such as CLIP learn aligned image-text embeddings that generalize across object categories and attributes [2]. Motivated by these results, prior work has explored conditioning embodied agents on natural language instructions to improve task understanding and generalization. Nevertheless, many approaches treat language as a high-level conditioning signal without explicitly grounding it in low-level spatial perception and action, limiting their effectiveness in precise navigation and manipulation tasks.

A central challenge in pixel-based embodied learning is the loss of spatial information in standard convolutional architectures. Common pooling and flattening operations discard explicit geometric structure, making it difficult for agents to localize small but task-critical objects. Prior work has shown that coordinate-based representations, such as Spatial Softmax, enable networks to extract explicit spatial keypoints from visual features, significantly improving visuomotor control from pixels [3]. However, the integration of such spatial abstractions with language-conditioned policies remains underexplored.

In this work, we present a Vision-Language-Action (VLA) agent designed to solve partially observable, sparse-reward navigation tasks in the MiniGrid environment [4]. Our approach combines a Spatial Softmax-based visual encoder with a frozen CLIP text encoder, fusing spatial and semantic representations within a recurrent policy trained using Proximal Policy Optimization (PPO) [5]. To address the exploration challenges posed by long-horizon tasks, we employ a multi-stage curriculum learning strategy [6], progressively transferring learned skills from simple navigation to complex multi-room environments.

Experimental results demonstrate that the proposed agent successfully transfers learned behaviors to a challenging 2D obstructed maze, achieving robust performance in settings where direct training fails to converge. Through systematic ablations, we show that both explicit spatial representations and curriculum-based training are essential for effective Vision-Language-Action learning from pixels.

II. RELATED WORK

A. Vision-Based Reinforcement Learning

Learning policies directly from pixel observations has been a long-standing goal in reinforcement learning. Deep RL methods have shown that convolutional neural networks can extract useful visual features for control, but such agents often exhibit poor sample efficiency and limited robustness in sparse-reward settings [1]. In partially observable environments, recurrent architectures such as LSTMs are commonly employed to aggregate temporal information and approximate belief states [7], though they remain sensitive to exploration failures.

B. Spatial Representations for Visuomotor Control

Standard CNN-based encoders frequently rely on global pooling or flattening, which remove explicit spatial structure. Levine et al. [3] introduced Spatial Softmax as a mechanism for extracting explicit coordinate representations from feature maps, enabling precise localization of task-relevant objects. Such spatial abstractions have been shown to be critical for pixel-based manipulation and navigation tasks, motivating their use in our visual encoder design.

C. Vision–Language Models for Embodied Agents

Vision–Language Models such as CLIP learn aligned semantic representations between images and text, enabling strong zero-shot generalization [2]. Recent work has explored using language embeddings to condition embodied agents on natural language goals or instructions. However, many approaches incorporate language at a high level without explicitly grounding it in spatial perception, limiting their effectiveness in tasks that require precise geometric reasoning.

D. Curriculum Learning in Sparse-Reward Environments

Curriculum learning has been widely adopted to mitigate exploration challenges in sparse-reward reinforcement learning. By organizing tasks from simple to complex, curricula can significantly improve convergence and transfer learning [6]. This strategy has proven particularly effective in navigation tasks with long dependency chains, aligning with our multi-phase training approach.

III. METHODOLOGY

We propose a Vision–Language–Action (VLA) architecture designed to solve partially observable, sparse-reward navigation tasks from high-dimensional pixel inputs. Our approach integrates a specialized visual encoder with a frozen language model and utilizes a multi-stage curriculum learning strategy to achieve convergence.

A. Problem Formulation

We formulate the task as a Partially Observable Markov Decision Process (POMDP) [1] defined by the tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \mathcal{O} \rangle$ within the MiniGrid environment suite [4]. The state space $\mathcal{S}$ represents the full grid configuration, which is hidden from the agent. At each time step t , the agent receives an observation $o_t \in \mathcal{O}$ consisting of a $56 \times 56 \times 3$ RGB image I_t representing a 7×7 ego-centric view, and a natural language goal instruction G (e.g., “find the blue ball”).

The action space $\mathcal{A}$ consists of 7 discrete actions: {turn left, turn right, move forward, pickup, drop, toggle, done}. The transition function $\mathcal{T}$ is deterministic but unknown. The reward function $\mathcal{R}$ is sparse, provided only upon successful task completion, necessitating the use of auxiliary reward shaping during training.

B. VLA Agent Architecture

Our agent architecture, illustrated in Fig. 1, consists of three primary modules: a perception module for vision and language, a fusion layer, and a recurrent policy network.

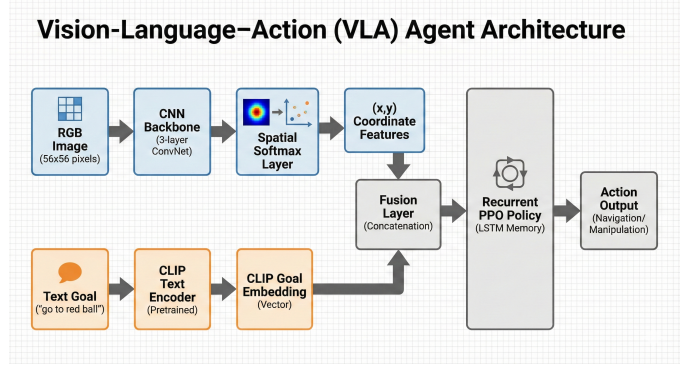


Fig. 1. The proposed Vision–Language–Action (VLA) architecture. The model processes ego-centric RGB images via a custom CNN with Spatial Softmax, fusing the output with CLIP-embedded goals before passing to a Recurrent PPO policy.

1) *Visual Encoder with Spatial Softmax*: To address the challenge of “physics blindness” in standard Convolutional Neural Networks (CNNs), we implement a custom visual encoder designed to extract explicit spatial coordinates from raw pixels. The encoder consists of a 3-layer CNN backbone with ReLU activations. Instead of using Global Average Pooling (GAP), which destroys spatial locality, we employ a *Spatial Softmax* layer [3].

For a feature map $F \in \mathbb{R}^{C \times H \times W}$, the Spatial Softmax operation computes the expected 2D coordinates (x_c, y_c) for each channel c :

$$x_c = \sum_{i,j} i \cdot \frac{e^{F_{c,i,j}}}{\sum_{i',j'} e^{F_{c,i',j'}}}, \quad y_c = \sum_{i,j} j \cdot \frac{e^{F_{c,i,j}}}{\sum_{i',j'} e^{F_{c,i',j'}}} \quad (1)$$

This mechanism forces the network to learn specific keypoints (e.g., the location of a key or door) relative to the agent’s position, outputting a compact coordinate vector $v_{img} \in \mathbb{R}^{2C}$.

2) *Language Embedding*: The textual goal G is encoded using a frozen CLIP (Contrastive Language–Image Pre-training) text encoder (ViT-B/32) [2]. We extract the semantic embedding $v_{lang} \in \mathbb{R}^{512}$ to ground the visual features in the task context. This allows the agent to generalize to different color/object combinations without retraining the language model.

3) *Recurrent Policy*: The visual vector v_{img} and language vector v_{lang} are concatenated and passed to a Recurrent Neural Network (RNN) utilizing Long Short-Term Memory (LSTM) cells [7]. The LSTM maintains a hidden state h_t to aggregate temporal information, effectively mapping the history of observations to a belief state. This is critical for solving the object permanence challenges in the *ObstructedMaze-2D* environment, where objects may be occluded or located in different rooms. The output of the LSTM feeds into separate Actor (policy) and Critic (value) heads trained via Proximal Policy Optimization (PPO) [5].

C. Curriculum Learning Strategy

Direct training on the target environment failed to converge due to the sparsity of rewards and the complexity of the

dependency chain (Search $\rightarrow$ Box $\rightarrow$ Key $\rightarrow$ Door $\rightarrow$ Ball). We therefore adopted a 4-phase curriculum learning approach [6]:

- 1) Phase 1: Navigation. The agent learns basic movement and collision avoidance in empty rooms.
- 2) Phase 2: Physics Manipulation. The agent learns to interact with objects (pickup key, toggle door) in a single room (*DoorKey*).
- 3) Phase 3: Logic & Object Permanence. The agent learns to open boxes to reveal hidden keys and unlock doors in a linear 1D layout (*ObstructedMaze-1D*).
- 4) Phase 4: Spatial Reasoning. The agent transfers learned skills to the complex 2D maze (*ObstructedMaze-2D*), integrating multi-room search strategies.

We utilize transfer learning between phases by initializing the agent’s weights with the best model from the previous phase, significantly reducing sample complexity.

D. Reward Shaping and Entropy Regularization

To guide exploration, we implemented a dynamic reward shaping mechanism. During early training phases, we provided dense ”breadcrumb” rewards: +0.3 for opening boxes, +0.3 for picking up keys, +1.0 for opening doors, and +10.0 for the final goal.

As the agent achieved proficiency, we introduced a ”starvation” protocol, annealing the intermediate rewards to 0.0 while maintaining the final goal reward. This forced the agent to optimize for efficiency and prevented local optima where the agent would ”farm” intermediate rewards without completing the task. We effectively balanced exploration and exploitation by dynamically tuning the entropy coefficient, starting at 0.01 to encourage random discovery and decaying to 0.0005 to enforce precise trajectory execution in the final model.

IV. RESULTS

We evaluated the performance of our VLA agent across the full curriculum and on the final *ObstructedMaze-2D* task using both rollout metrics and deterministic evaluation episodes.

A. Curriculum Training Progression

Fig. 2 illustrates the agent’s cumulative training progress across all four curriculum phases. The x-axis represents the total number of time steps (approx. 25 million) aggregated across the entire learning lifecycle.

- Skill Acquisition (0–14M steps): During the initial phases (Navigation, Manipulation, Logic), the mean reward remains relatively low. This reflects the agent learning low-level primitives (e.g., collision avoidance, door toggling) where rewards are sparse or lower in magnitude compared to the final objective.
- Skill Transfer & Mastery (14M–25M steps): Upon transitioning to the final spatial reasoning phase, we observe a sharp inflection point. The mean rollout reward rises rapidly from ~ 1.0 to a peak of 6.37. This indicates that the agent successfully transferred its primitive skills to solve the complex dependency chain of the 2D maze.

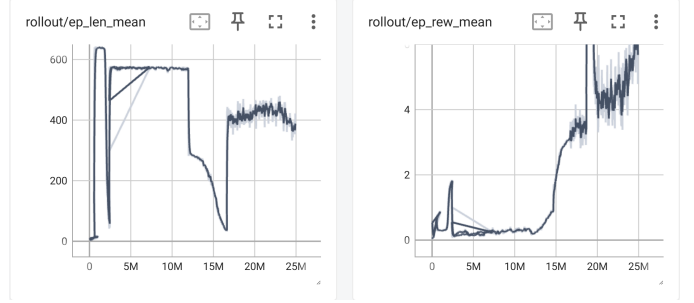


Fig. 2. Training results on *ObstructedMaze-2D*. **(Left)** Mean Rollout Reward shows a sharp rise to 6.37 after the curriculum transition (14M steps), indicating successful mastery of the task. **(Right)** Mean Episode Length drops significantly from the timeout limit (576) to ~ 380 steps, confirming that the agent is solving the maze efficiently rather than relying on timeouts.

The concurrent drop in episode length (Fig. 2, Left) from the timeout limit (576) to ~ 380 steps confirms that the high reward corresponds to efficient task completion rather than accidental discovery.

B. Evaluation and Robustness

We conducted periodic evaluations using a deterministic policy. The evaluation reward fluctuated (Mean ≈ 0.34) despite the high training success rate (6.37). This discrepancy highlights the ”Deterministic Trap” in sparse-reward navigation: while the stochastic training policy ($\alpha = 0.001$) retains sufficient noise to navigate out of local optima, the deterministic evaluation policy is brittle on difficult procedural seeds. Visual verification using stochastic sampling confirmed that the agent retains the ability to solve the maze across diverse layouts.

C. Ablation Analysis

To validate our specific design choices, we conducted two key ablation studies focusing on the training strategy and the network architecture.

- 1) Impact of Spatial Softmax: To assess the necessity of our visual encoder, we replaced the Spatial Softmax layer with a standard Flatten operation (common in basic RL baselines). We observed that this baseline failed to solve even the simpler Phase 2 task (*DoorKey*). While the standard CNN could learn basic navigation, it failed to learn the precise alignment required for interaction (e.g., facing the door exactly to toggle it). This demonstrates that explicit coordinate extraction (x, y) is essential for pixel-based manipulation tasks where target objects occupy minimal screen space.
- 2) Impact of Curriculum Learning: We attempted to train the agent directly on the final *ObstructedMaze-2D* task without the 4-phase curriculum. In these experiments, the agent failed to acquire any meaningful behavior (Mean Reward ≈ 0.0), effectively getting stuck in the initial room. This confirms that decomposing the problem into progressively harder phases is critical for pixel-based PPO agents to overcome the exploration bottleneck.

V. CONCLUSION

This work presents a Vision–Language–Action agent capable of solving partially observable, sparse-reward navigation tasks directly from pixel observations. By combining spatially grounded visual representations with language-conditioned goal embeddings and a structured curriculum learning strategy, we demonstrate that long-horizon reasoning can be learned effectively in challenging grid-based environments.

Our results highlight two key findings. First, explicit spatial representations derived via Spatial Softmax are critical for precise interaction and object localization, particularly in settings where task-relevant objects occupy minimal visual space. Second, curriculum learning with staged transfer is essential for overcoming the exploration bottleneck inherent in sparse-reward environments, enabling successful policy learning where direct end-to-end training fails.

While the proposed approach achieves strong performance in the MiniGrid ObstructedMaze environment, several limitations remain. The reliance on hand-designed curricula and reward shaping may reduce scalability to more complex or unstructured domains. Additionally, the observed discrepancy between stochastic training performance and deterministic evaluation underscores the brittleness of learned policies under strict execution constraints.

Future work will focus on reducing manual curriculum design, incorporating adaptive or self-paced learning strategies, and extending the framework to more realistic simulation platforms and continuous control tasks. We also aim to explore tighter integration between language reasoning and low-level control, enabling agents to better leverage semantic structure for efficient exploration and generalization. We believe this work provides a principled step toward robust, interpretable Vision–Language–Action learning in embodied environments.

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